Requirement Gathering – AI-Based Environmental Compliance Monitoring and Rule-Based Risk Prediction System

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1. Project Overview

The AI-Based Environmental Compliance Monitoring and Rule-Based Risk Prediction System is a web-based application developed to assist industries in monitoring environmental compliance efficiently and accurately. The system is designed to replace traditional manual monitoring methods by providing a centralized digital platform for recording, managing, and analyzing environmental data. It enables industries to monitor important environmental parameters such as Air Quality Index (AQI), water quality (pH level), temperature, humidity, waste disposal status, and inspection details.

The system compares the collected environmental data with predefined environmental standards and regulatory limits. Using rule-based IF–THEN logic, it automatically predicts the compliance risk level as Low, Medium, or High and provides suitable recommendations for corrective actions. This helps industries identify potential environmental violations at an early stage and supports regulatory compliance. The system also generates reports and dashboards that assist management and environmental authorities in making informed decisions.

2. System Scope

This project is developed as an MCA Mini Project focusing on digital environmental compliance management. The system includes user authentication, industry management, environmental monitoring, compliance verification, rule-based risk prediction, dashboard visualization, and report generation.

The application allows environmental inspectors to enter inspection data, administrators to manage environmental standards, and industry managers to monitor compliance status through a centralized dashboard. The system maintains historical environmental records, making it easier to analyze previous inspections and identify recurring compliance issues.

The future scope of the project includes integrating IoT sensors for automatic environmental data collection, Machine Learning algorithms for intelligent risk prediction, GIS-based pollution visualization, automated SMS and email notifications, and mobile application support for field inspectors.

3. Target Audience

The system is intended for organizations and industries that are required to monitor environmental compliance regularly.

The primary users include:

- Environmental Inspectors responsible for conducting inspections and recording environmental readings.
- Industry Managers who monitor compliance reports and ensure environmental regulations are followed.
- Factory Supervisors who oversee daily environmental monitoring activities.
- Environmental Officers who verify compliance and recommend corrective actions.
- Regulatory Authorities such as Pollution Control Boards that review environmental reports submitted by industries.

Each user accesses different modules according to their responsibilities through role-based access control.

4. Modules

The system is divided into several functional modules to ensure proper management of environmental compliance activities.

User Authentication Module

Provides secure login, logout, password management, and role-based access for different users.

Industry Management Module

Maintains complete details of industries, factories, locations, contact information, and environmental categories.

Environmental Monitoring Module

Allows inspectors to enter environmental parameters including Air Quality Index (AQI), water quality (pH), temperature, humidity, waste disposal status, and inspection remarks.

Compliance Monitoring Module

Compares recorded environmental values with predefined environmental standards and determines whether the industry complies with environmental regulations.

Rule-Based Risk Prediction Module

Uses predefined IF–THEN rules to classify compliance risk as Low, Medium, or High based on recorded environmental data.

Dashboard Module

Displays real-time information such as total inspections, compliance percentage, active environmental alerts, risk levels, and environmental trends using charts and graphs.

Report Management Module

Generates inspection reports, compliance reports, violation reports, environmental summaries, and historical reports for management and regulatory authorities.

5. User Roles

The system supports different user roles with specific responsibilities.

Administrator

- Manages user accounts.
- Maintains industry details.
- Defines environmental compliance standards.
- Updates system settings.
- Generates system-wide reports.
- Monitors overall environmental compliance.

Environmental Inspector

- Conducts environmental inspections.
- Records environmental readings.
- Updates inspection observations.
- Verifies compliance status.
- Reviews generated alerts.

Industry Manager

- Views inspection reports.

- Monitors compliance status.
- Reviews risk prediction results.
- Implements corrective actions.
- Tracks environmental performance.

Each user has different access permissions to ensure data security and prevent unauthorized modifications.

6. System Ownership

The AI-Based Environmental Compliance Monitoring and Rule-Based Risk Prediction System is developed as an academic Mini Project under the Integrated MCA programme. The project is intended for educational purposes to demonstrate the application of Artificial Intelligence concepts, rule-based systems, and environmental monitoring in software development.

The system can later be enhanced and deployed for real industrial applications by integrating IoT sensors, cloud computing, predictive analytics, and machine learning technologies.

7. Industry / Domain

The project belongs to the **Environmental Monitoring and Compliance Management** domain.

It is suitable for industries including:

- Chemical Industries
- Pharmaceutical Industries
- Textile Industries
- Cement Industries
- Food Processing Industries
- Steel Manufacturing Industries
- Paper Industries
- Plastic Manufacturing Industries

- Power Plants
- Waste Management Industries

The system can also be adopted by government organizations responsible for environmental monitoring and pollution control.

8. Data Collection Contact

Name: Environmental Officer / Factory Supervisor

Role: Responsible for monitoring environmental compliance, maintaining inspection records, supervising pollution control activities, and coordinating with regulatory authorities.

Organization: Industrial Manufacturing Unit / Pollution Control Department

Contact Details: To be filled based on the organization visited during requirement gathering.

9. Questionnaire for data collection

Q1. How do you currently monitor environmental compliance?

Answer: Environmental compliance is monitored through manual inspections, laboratory testing, sensor measurements, and periodic verification of environmental records.

Q2. Which environmental parameters are monitored regularly?

Answer: Air Quality Index (AQI), water quality (pH), temperature, humidity, emissions, waste disposal activities, and pollution levels are monitored.

Q3. How frequently are inspections conducted?

Answer: Inspections are generally conducted weekly, monthly, quarterly, or immediately when complaints or environmental incidents occur.

Q4. What problems are faced in the existing system?

Answer: Manual record keeping, delayed reporting, human errors, scattered information, lack of real-time monitoring, and difficulty in identifying compliance violations.

Q5. How are environmental records maintained?

Answer: Most industries use spreadsheets, paper registers, laboratory reports, and manually prepared inspection reports.

Q6. Would real-time monitoring improve environmental management?

Answer: Yes. Real-time monitoring provides immediate environmental information, reduces manual work, improves accuracy, and enables faster corrective actions.

Q7. What are the common environmental violations observed?

Answer: High air pollution, improper waste disposal, water contamination, excessive emissions, and delayed corrective actions are the most common violations.

Q8. Would automatic compliance checking be useful?

Answer: Yes. Automatic compliance checking reduces human effort, minimizes errors, and quickly identifies environmental violations.

Q9. What reports are required?

Answer: Environmental inspection reports, compliance reports, pollution reports, violation reports, corrective action reports, and historical environmental reports.

Q10. What future features would improve the system?

Answer: IoT-based automatic environmental monitoring, Machine Learning-based pollution prediction, mobile applications, GIS-based pollution mapping, automated notifications, cloud storage, and predictive analytics dashboards.

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